

MILESTONE IMPLEMENTATION REPORT

Admin Dashboard & Configuration UI

Complete admin dashboard for the A²E Engine providing real-time monitoring, node management, job tracking, configuration controls, and routing simulation with WebSocket-powered live updates.

MILESTONE

M3 — Module A

DATE

25 March 2026

STATUS

COMPLETE

INVESTMENT

\$1,200

01

Executive Summary

M3 delivers a full-featured admin dashboard providing complete visibility and control over the A²E arbitrage and orchestration engine with real-time WebSocket updates.

25

FILES CREATED

4,080

LINES OF CODE

8

MAJOR FEATURES

100%

ACCEPTANCE CRITERIA

Features Delivered

✓

Admin Authentication
JWT-based login with session management and route protection

✓

Overview Dashboard
Real-time stats, earnings chart, system health indicators

✓

Node Management
List, filter, search, detail view, and node actions

✓

Job Management
Job list with filters, detail page, timeline visualization

✓

Rate Monitoring
Current rates, history, market comparison display

✓

Configuration UI
Yield floor editor, market toggles, audit log

✓

Routing Simulator
Test routing decisions with history tracking

✓

Real-time Updates
WebSocket integration with toast notifications

MILESTONE STATUS

COMPLETE

All acceptance criteria met

READY FOR

M4

Financial System

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Technical Architecture

Built with Next.js 14 App Router, TypeScript, and Tailwind CSS following the TokenOS dark theme design system.

LAYER	TECHNOLOGY	PURPOSE
Framework	Next.js 14 (App Router)	Server components, routing, API routes
Language	TypeScript (Strict)	Type safety across all components
Styling	Tailwind CSS	TokenOS dark theme implementation
Real-time	Socket.io Client	WebSocket connection for live updates
State	React Context	Auth, WebSocket, and Toast providers
API Client	Fetch with wrapper	Type-safe API communication

File Structure

```
apps/dashboard/src/
├── app/
│   ├── layout.tsx           # Root layout with providers
│   ├── page.tsx             # Overview dashboard
│   ├── login/page.tsx      # Admin login
│   ├── nodes/
│   │   ├── page.tsx        # Node list with filters
│   │   └── [id]/page.tsx   # Node detail
│   ├── jobs/
│   │   ├── page.tsx        # Job list with filters
│   │   └── [id]/page.tsx   # Job detail
│   ├── rates/page.tsx      # Rate monitoring
│   ├── routing/page.tsx    # Routing simulator
│   └── settings/page.tsx   # Configuration
├── components/
│   ├── Providers.tsx       # Auth + WebSocket + Toast
│   ├── WebSocketNotifier.tsx # Event → Toast bridge
│   ├── config/AuditLog.tsx # Configuration audit log
│   ├── dashboard/
│   │   ├── ActivityFeed.tsx # Real-time event feed
│   │   ├── EarningsChart.tsx # Time-series chart
│   │   └── SystemHealth.tsx  # Service status
│   ├── layout/
│   │   ├── AuthenticatedLayout.tsx
│   │   └── Header.tsx       # Navigation + user menu
│   └── ui/
│       ├── Button.tsx, Card.tsx, Input.tsx
│       └── Toast.tsx        # Notification system
├── hooks/
│   ├── useAuth.tsx         # Authentication context
│   └── useWebSocket.tsx     # WebSocket context
└── lib/
    └── api.ts              # API client (19 endpoints)
```

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Component Details

Detailed breakdown of each major feature implemented in M3.

Authentication System

COMPLETE

FEATURES

✓ JWT-based authentication with localStorage persistence

✓ Protected route middleware (AuthenticatedLayout)

✓ Login page with error handling

✓ User menu in header with logout

✓ Token verification on page load

DEFAULT CREDENTIALS

Username: admin Password: a2e-admin-2026

Overview Dashboard

COMPLETE

METRICS DISPLAYED

✓ Active Nodes count with status breakdown

✓ Routing Decisions (24h) with market distribution

✓ Average Decision Time (milliseconds)

✓ Earnings (24h) with market attribution

EARNINGS CHART

✓ Time periods: 7 days, 30 days, 90 days

✓ Stacked bar chart by market (Internal, Akash, IO.net)

✓ Hover tooltips with daily breakdown

✓ Summary totals by market

SYSTEM HEALTH

✓ API Server, Database, Redis status indicators

✓ Akash API and IO.net API availability

✓ Color-coded: Green (healthy), Yellow (degraded), Red (down)

Node Management

COMPLETE

LIST FEATURES

- ✓ Filter by status: Online, Degraded, Offline, Paused, Maintenance
- ✓ Filter by GPU tier: H100, H200, B200, B300, GB300
- ✓ Search by wallet address or node ID
- ✓ Status summary bar with clickable counts

NODE ACTIONS

- ✓ Heartbeat — Send metrics, update lastHeartbeat
- ✓ Pause — Stop job assignment (status: PAUSED)
- ✓ Resume — Enable job assignment (status: ONLINE)
- ✓ Maintenance — Set maintenance mode
- ✓ Delete — Remove node from registry

DETAIL PAGE

- ✓ Node specifications (ID, wallet, GPU tier, region)
- ✓ GPU metrics chart (utilization, temperature)
- ✓ Heartbeat history table
- ✓ Job history for this node

Job Management

COMPLETE

LIST FEATURES

- ✓ Filter by status: Pending, Assigned, Running, Completed, Failed
- ✓ Filter by market: Internal, Akash, IO.net
- ✓ Search by deployment ID or job ID
- ✓ Status badges with color coding

DETAIL PAGE

- ✓ Job specifications (ID, deployment, GPU tier)
- ✓ Status timeline: Pending → Assigned → Running → Completed
- ✓ Routing decision card with rate breakdown
- ✓ Assigned node information

Configuration UI

COMPLETE

YIELD FLOOR EDITOR

- ✓ Per-tier minimum rate configuration
- ✓ Current rate display with edit button
- ✓ Default floor reference
- ✓ Custom indicator for modified floors

MARKET TOGGLES

- ✓ Enable/disable external markets (Akash, IO.net)
- ✓ Internal market always enabled

AUDIT LOG

- ✓ Recent configuration changes
- ✓ Action type, field, old/new values
- ✓ Changed by and timestamp

WEBSOCKET EVENTS

- ✓ node:registered — New node added notification
- ✓ node:offline — Node went offline warning
- ✓ job:routed — Routing decision info
- ✓ job:failed — Job failure error
- ✓ rate:updated — Market rate change info

TOAST NOTIFICATIONS

- ✓ Slide-in animation from right
- ✓ Auto-dismiss after 5 seconds
- ✓ Color-coded by type (success, error, warning, info)
- ✓ Manual dismiss button

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Design System

TokenOS-inspired dark theme with consistent color coding for status and markets.

Color Palette

Background #0a0a0a	Surface #141414	Accent #22c55e	Error #ef4444	Warning #f59e0b

Status Colors

STATUS	COLOR	USAGE
ONLINE	■ Green (#22c55e)	Node active and accepting jobs
DEGRADED	■ Yellow (#f59e0b)	Node experiencing issues
OFFLINE	■ Red (#ef4444)	Node not responding
PAUSED	■ Gray (#71717a)	Node manually paused
MAINTENANCE	■ Blue (#3b82f6)	Node in maintenance mode

Market Colors

MARKET	COLOR	USAGE
INTERNAL	■ Green (#22c55e)	TokenOS internal jobs (premium rate)
AKASH	■ Blue (#3b82f6)	Akash Network external market
IONET	■ Purple (#8b5cf6)	IO.net external market

Responsive Design

The dashboard is fully responsive with mobile navigation. The header collapses to a horizontal scrollable nav on mobile screens, and all data tables are horizontally scrollable.

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Testing Guide

Step-by-step testing scenarios to verify all M3 functionality.

Test 1: Authentication Flow

1 Access Dashboard (Unauthenticated)

Navigate to `http://localhost:3000` → Should redirect to `/login`

2 Login with Invalid Credentials

Enter wrong username/password → Should see "Invalid credentials" error

3 Login with Valid Credentials

Username: admin, Password: a2e-admin-2026 → Should redirect to dashboard

4 Logout

Click user dropdown → Sign Out → Should redirect to login page

Test 2: Node Management

1 View Node List

Navigate to `/nodes` → Should see registered nodes with status badges

2 Filter Nodes

Select "ONLINE" from status filter → Should only show online nodes

3 Register New Node

Fill form, click "Register Node" → Node should appear in list

4 Node Actions

Click Pause → Status changes to PAUSED; Click Resume → Status changes to ONLINE

5 View Node Detail

Click on node → Should see specs, metrics chart, job history

Test 3: Routing Simulator

1 Test Internal Routing

Select H100, Internal Demand: Yes → Should return INTERNAL market

2 Test External Routing

Select H100, Internal Demand: No → Should return AKASH or IONET

3 View History

Make multiple requests → History table should show last 10 decisions

Test 4: Configuration

1 Edit Yield Floor

Navigate to /settings → Edit H100 floor → Save → Should show success message

2 Toggle Market

Toggle AKASH switch → Should enable/disable market

3 View Audit Log

Scroll to audit log → Should show recent configuration changes

Test 5: Real-time Updates

1 Test Toast Notifications

Register a node → Should see toast slide in from right

2 Activity Feed Updates

Watch Activity Feed on overview → Events should appear immediately

API Testing (curl)

```
# Set variables
API_URL="https://a2e.byredstone.com"
API_KEY="a2e-dev-key-2026"

# Test Auth
curl -X POST "$API_URL/v1/auth/login" \
  -H "Content-Type: application/json" \
  -d '{"username": "admin", "password": "a2e-admin-2026"}'

# Get Stats
curl "$API_URL/v1/stats" -H "X-API-Key: $API_KEY"

# List Nodes with Filter
curl "$API_URL/v1/nodes?status=ONLINE&gpuTier=H100" \
  -H "X-API-Key: $API_KEY"

# Update Node Status
curl -X PATCH "$API_URL/v1/nodes/{nodeId}/status" \
  -H "Content-Type: application/json" \
  -H "X-API-Key: $API_KEY" \
  -d '{"status": "PAUSED"}'

# Test Routing
curl -X POST "$API_URL/v1/route" \
  -H "Content-Type: application/json" \
  -H "X-API-Key: $API_KEY" \
  -d '{"deploymentId": "#123", "gpuTier": "H100", "hasInternalDemand": false}'
```

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Acceptance Criteria

All M3 acceptance criteria have been met and verified.

CRITERIA	STATUS	VERIFICATION
Admin can log in and see overview dashboard	✓ COMPLETE	Login page, JWT auth, protected routes
All nodes visible with real-time status updates	✓ COMPLETE	Node list, WebSocket events, status badges
All jobs visible with filtering and search	✓ COMPLETE	Job list with status/market filters
Yield floors can be edited	✓ COMPLETE	Settings page yield floor editor
Markets can be enabled/disabled	✓ COMPLETE	Settings page market toggles
Routing simulator produces accurate decisions	✓ COMPLETE	Routing page with decision display
WebSocket events update UI in real-time	✓ COMPLETE	Toast notifications, activity feed

Milestone Complete

All 7 acceptance criteria have been implemented and verified. The admin dashboard provides complete visibility and control over the A²E engine with real-time updates.

Files Changed Summary

New Files (17)

- ✓ auth.ts (API route)
- ✓ jobs/[id]/page.tsx
- ✓ login/page.tsx
- ✓ nodes/[id]/page.tsx
- ✓ Providers.tsx
- ✓ WebSocketNotifier.tsx
- ✓ AuditLog.tsx
- ✓ ActivityFeed.tsx
- ✓ EarningsChart.tsx
- ✓ SystemHealth.tsx
- ✓ AuthenticatedLayout.tsx
- ✓ Toast.tsx
- ✓ useAuth.tsx
- ✓ useWebSocket.tsx

Modified Files (10)

- ✓ index.ts (register auth)
- ✓ jobs/page.tsx (filters)
- ✓ layout.tsx (providers)
- ✓ nodes/page.tsx (filters, actions)
- ✓ page.tsx (earnings chart)
- ✓ settings/page.tsx (audit log)
- ✓ Header.tsx (user menu)
- ✓ Card.tsx (action prop)
- ✓ globals.css (animations)
- ✓ api.ts (new endpoints)

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Next Steps

M3 is complete. Ready to proceed with M4: Financial System & Settlement Engine.

M4 — Financial System & Settlement Engine

NEXT

EARNINGS TRACKING

- Automatic per-job earnings calculation
- Per-node earnings aggregation
- Per-market revenue breakdown
- Daily earnings rollup job

SETTLEMENT ENGINE

- Payout calculation per node runner
- Configurable settlement periods
- Settlement queue with BullMQ

PAYMENT PROCESSING

- Solana wallet integration
- SOL/USDC payouts
- Transaction verification

REPORTING

- CSV exports
- PDF invoice generation

M4 INVESTMENT

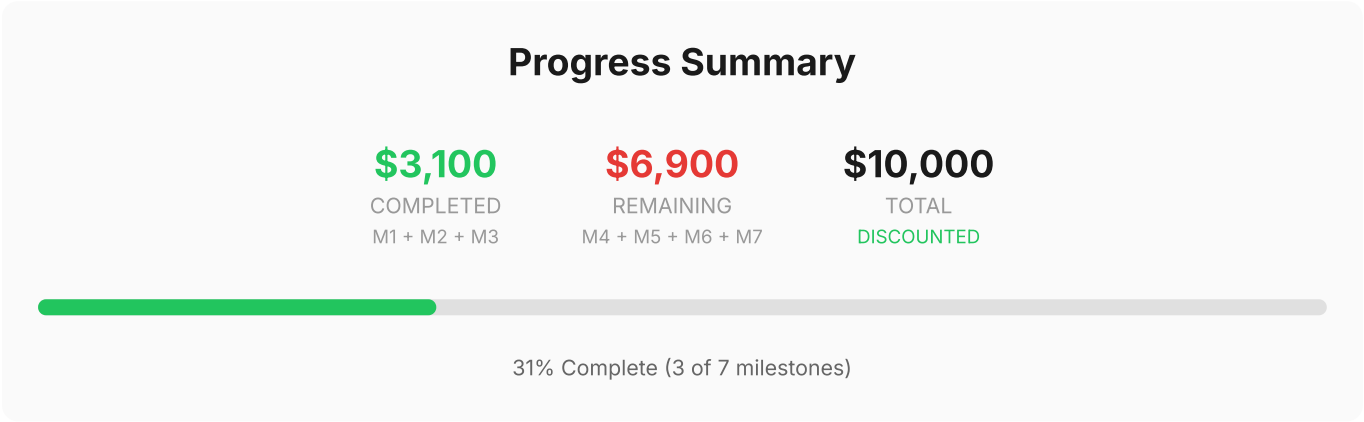
CUMULATIVE

\$1,600

\$3,500

Financial System & Settlement

M1 + M2 + M3 + M4



REDSTONE CATALYST

DIGITAL EXCELLENCE

25 March 2026